Grokking the Coding Interview: Patterns for Coding Questions

Fast track your coding interview prep by learning coding patterns. Act smartly, and follow the coding patterns.

Level: Intermediate | Study Time: 321h | Lessons: 480 | Playgrounds : 517

2,11,991 (Students)

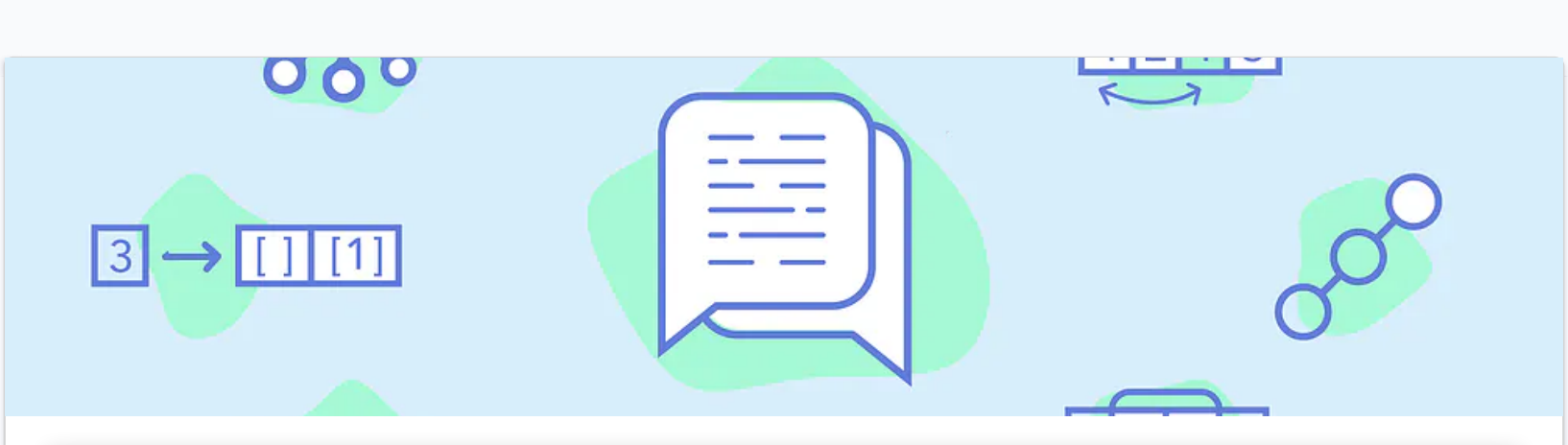
Course Overview

Coding interviews are getting harder every day. A few years back, brushing up on key data structures and going through 50-75 coding interview questions was more than enough prep for an interview. Today, everyone has access to massive sets of coding problems, and they have gotten more difficult to account for that. The process has gotten more competitive.

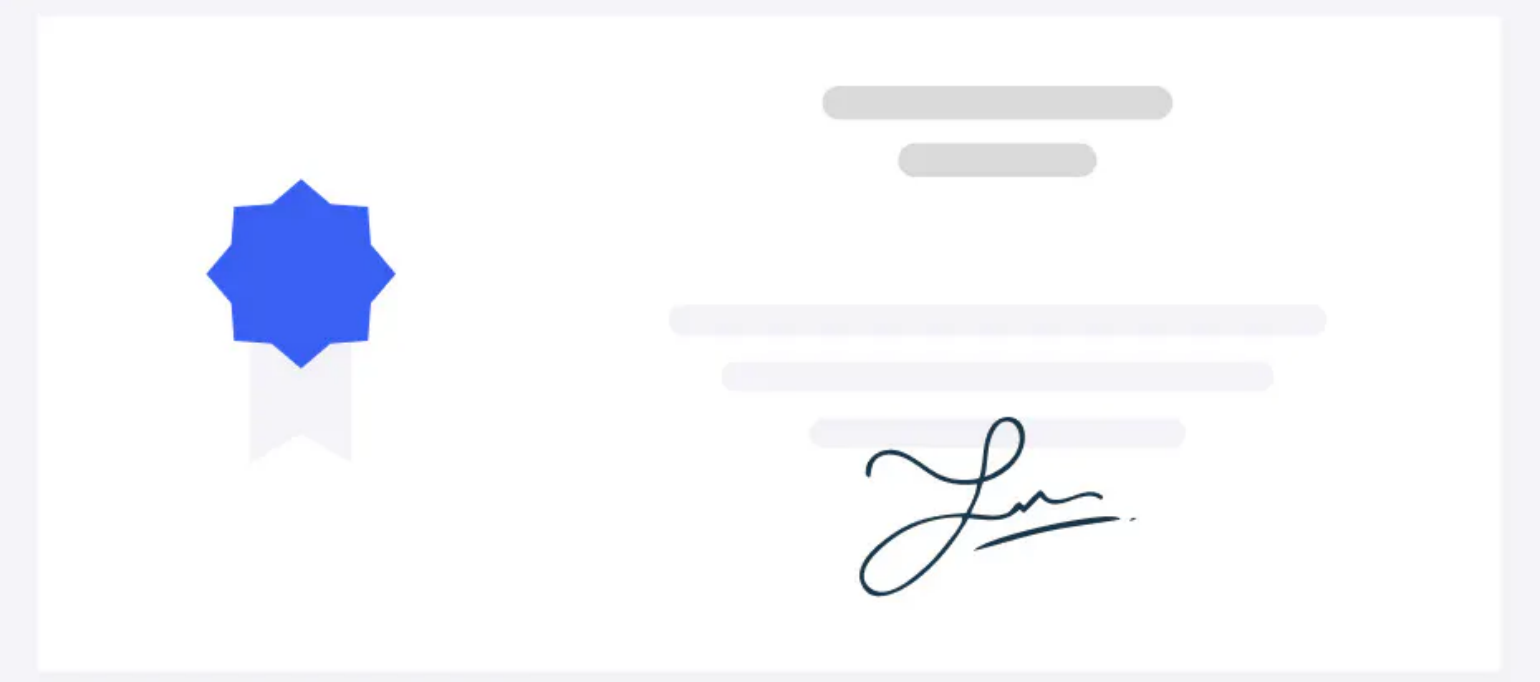
When our team sat together to brainstorm on ideas to make the interview process easier for candidates, we realized quickly that one skill helped us the most when we were preparing for coding interviews: "the ability to map a new problem to an already known problem."

To help candidates with that, we have come up with a list of 27 common coding patterns to help solve coding questions. The idea is, once you are familiar with a pattern, you'll be able to solve dozens of problems with it.

The techniques taught in this course have helped developers land jobs in top companies including Google, Facebook, Amazon, and Microsoft.



Start Reading



Certification on completing the course.

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Course Content

1. Introduction
2 chapters

- Who Should Take This Course?
- Course Overview

2. Warmup
17 chapters

- Why warmup?
- Contains Duplicate (easy)
- Solution: Contains Duplicate
- Pangram (easy)
- Solution: Pangram
- Reverse Vowels (easy)
- Solution: Reverse Vowels
- Valid Palindrome (easy)
- Solution: Valid Palindrome
- Valid Anagram (easy)
- Solution: Valid Anagram
- Shortest Word Distance (easy)
- Solution: Shortest Word Distance
- Number of Good Pairs (easy)
- Solution: Number of Good Pairs
- Sqrt (medium)
- Solution: Sqrt

3. Pattern: Two Pointers
23 chapters

- Introduction to Two Pointers Pattern
- Pair with Target Sum (easy)
- Solution: Pair with Target Sum
- Find Non-Duplicate Number Instances (easy)
- Solution: Find Non-Duplicate Number Instances
- Squaring a Sorted Array (easy)
- Solution: Squaring a Sorted Array
- Triplet Sum to Zero (medium)
- Solution: Triplet Sum to Zero
- Triplet Sum Close to Target (medium)
- Solution: Triplet Sum Close to Target
- Triplets with Smaller Sum (medium)
- Solution: Triplets with Smaller Sum
- Subarrays with Product Less than a Target (medium)
- Solution: Subarrays with Product Less than a Target
- Dutch National Flag Problem (medium)
- Solution: Dutch National Flag Problem
- Problem Challenge 1: Quadruple Sum to Target (medium)
- Solution: Problem Challenge 1: Quadruple Sum to Target
- Problem Challenge 2: Comparing Strings containing Backspaces (medium)
- Solution: Problem Challenge 2: Comparing Strings containing Backspaces
- Problem Challenge 3: Minimum Window Sort (medium)
- Solution: Problem Challenge 3: Minimum Window Sort

4. Pattern: Fast & Slow Pointers
15 chapters

- Introduction to Fast & Slow Pointers Pattern
- LinkedList Cycle (easy)
- Solution: LinkedList Cycle
- Middle of the LinkedList (easy)
- Solution: Middle of the LinkedList
- Start of LinkedList Cycle (medium)
- Solution: Start of LinkedList Cycle
- Happy Number (medium)
- Solution: Happy Number
- Problem Challenge 1: Palindrome LinkedList (medium)
- Solution: Problem Challenge 1: Palindrome LinkedList
- Problem Challenge 2: Rearrange a LinkedList (medium)
- Solution: Problem Challenge 2: Rearrange a LinkedList
- Problem Challenge 3: Cycle in a Circular Array (hard)
- Solution: Problem Challenge 3: Cycle in a Circular Array

5. Pattern: Sliding Window
21 chapters

- Introduction to Sliding Window Pattern
- Maximum Sum Subarray of Size K (easy)
- Solution: Maximum Sum Subarray of Size K
- Smallest Subarray With a Greater Sum (easy)
- Solution: Smallest Subarray With a Greater Sum
- Longest Substring with K Distinct Characters (medium)
- Solution: Longest Substring with K Distinct Characters
- Fruits into Baskets (medium)
- Solution: Fruits into Baskets
- Longest Substring with Same Letters after Replacement (hard)
- Solution: Longest Substring with Same Letters after Replacement
- Longest Subarray with Ones after Replacement (hard)
- Solution: Longest Subarray with Ones after Replacement
- Problem Challenge 1: Permutation in a String (hard)
- Solution: Problem Challenge 1: Permutation in a String
- Problem Challenge 2: String Anagrams (hard)
- Solution: Problem Challenge 2: String Anagrams
- Problem Challenge 3: Smallest Window containing Substring (hard)
- Solution: Problem Challenge 3: Smallest Window containing Substring
- Problem Challenge 4: Words Concatenation (hard)
- Solution: Problem Challenge 4: Words Concatenation

6. Pattern: Merge Intervals
15 chapters

- Introduction to Merge Intervals Pattern
- Merge Intervals (medium)
- Solution: Merge Intervals
- Insert Interval (medium)
- Solution: Insert Interval
- Intervals Intersection (medium)
- Solution: Intervals Intersection
- Conflicting Appointments (medium)
- Solution: Conflicting Appointments
- Problem Challenge 1: Minimum Meeting Rooms (hard)
- Solution: Problem Challenge 1: Minimum Meeting Rooms
- Problem Challenge 2: Maximum CPU Load (hard)
- Solution: Problem Challenge 2: Maximum CPU Load
- Problem Challenge 3: Employee Free Time (hard)
- Solution: Problem Challenge 3: Employee Free Time

7. Pattern: Cyclic Sort
17 chapters

- Introduction to Cyclic Sort Pattern
- Cyclic Sort (easy)
- Solution: Cyclic Sort
- Find the Missing Number (easy)
- Solution: Find the Missing Number
- Find all Missing Numbers (easy)
- Solution: Find all Missing Numbers
- Find the Duplicate Number (easy)
- Solution: Find the Duplicate Number
- Find all Duplicate Numbers (easy)
- Solution: Find all Duplicate Numbers
- Problem Challenge 1: Find the Corrupt Pair (easy)
- Solution: Problem Challenge 1: Find the Corrupt Pair
- Problem Challenge 2: Find the Smallest Missing Positive Number (medium)
- Solution: Problem Challenge 2: Find the Smallest Missing Positive Number
- Problem Challenge 3: Find the First K Missing Positive Numbers (hard)
- Solution: Problem Challenge 3: Find the First K Missing Positive Numbers

8. Pattern: In-place Reversal of a Linked List
11 chapters

- Introduction to In-place Reversal of a Linked List Pattern
- Reverse a LinkedList (easy)
- Solution: Reverse a LinkedList
- Reverse a Sub-list (medium)
- Solution: Reverse a Sub-list
- Reverse every K-element Sub-list (medium)
- Solution: Reverse every K-element Sub-list
- Problem Challenge 1: Reverse alternating K-element Sub-list (medium)
- Solution: Problem Challenge 1: Reverse alternating K-element Sub-list
- Problem Challenge 2: Rotate a LinkedList (medium)
- Solution: Problem Challenge 2: Rotate a LinkedList

9. Pattern: Stacks
16 chapters

- Introduction to Stack
- Operations on Stack
- Implementing Stack Data Structure
- Applications of Stack
- Problem 1: Balanced Parentheses
- Solution: Balanced Parentheses
- Problem 2: Reverse a String
- Solution: Reverse a String
- Problem 3: Decimal to Binary Conversion
- Solution: Decimal to Binary Conversion
- Problem 4: Next Greater Element
- Solution: Next Greater Element
- Problem 5: Sorting a Stack
- Solution: Sorting a Stack
- Problem 6: Simplify Path
- Solution: Simplify Path

10. Pattern: Monotonic Stack (*NEW*)
13 chapters

- Introduction to Monotonic Stack
- Next Greater Element (easy)
- Solution: Next Greater Element
- Daily Temperatures (easy)
- Solution: Daily Temperatures
- Remove Nodes From Linked List (easy)
- Solution: Remove Nodes From Linked List
- Remove All Adjacent Duplicates In String (easy)
- Solution: Remove All Adjacent Duplicates In String
- Remove All Adjacent Duplicates in String II (medium)
- Solution: Remove All Adjacent Duplicates in String II
- Remove K Digits (hard)
- Solution: Remove K Digits

11. Pattern: Hash Maps
13 chapters

- Introduction to Hashing
- Introduction to Hash Tables
- Issues with Hash Tables
- Problem 1: First Non-repeating Character (easy)
- Solution: First Non-repeating Character (easy)
- Problem 2: Largest Unique Number (easy)
- Solution: Largest Unique Number (easy)
- Problem 3: Maximum Number of Balloons (easy)
- Solution: Maximum Number of Balloons (easy)
- Problem 4: Longest Palindrome(easy)
- Solution: Longest Palindrome(easy)
- Problem 5: Ransom Note (easy)
- Solution: Ransom Note (easy)

12. Pattern: Tree Breadth First Search
19 chapters

- Introduction to Tree Breadth First Search Pattern
- Binary Tree Level Order Traversal (easy)
- Solution: Binary Tree Level Order Traversal
- Reverse Level Order Traversal (easy)
- Solution: Reverse Level Order Traversal
- Zigzag Traversal (medium)
- Solution: Zigzag Traversal
- Level Averages in a Binary Tree (easy)
- Solution: Level Averages in a Binary Tree
- Minimum Depth of a Binary Tree (easy)
- Solution: Minimum Depth of a Binary Tree
- Level Order Successor (easy)
- Solution: Level Order Successor
- Connect Level Order Siblings (medium)
- Solution: Connect Level Order Siblings
- Problem Challenge 1: Connect All Level Order Siblings (medium)
- Solution: Problem Challenge 1: Connect All Level Order Siblings
- Problem Challenge 2: Right View of a Binary Tree (easy)
- Solution: Problem Challenge 2: Right View of a Binary Tree

13. Pattern: Tree Depth First Search
15 chapters

- Introduction to Tree Depth First Search Pattern
- Binary Tree Path Sum (easy)
- Solution: Binary Tree Path Sum
- All Paths for a Sum (medium)
- Solution: All Paths for a Sum
- Sum of Path Numbers (medium)
- Solution: Sum of Path Numbers
- Path With Given Sequence (medium)
- Count Paths for a Sum (medium)
- Solution: Count Paths for a Sum
- Problem Challenge 1: Tree Diameter (medium)
- Solution: Problem Challenge 1: Tree Diameter
- Problem Challenge 2: Path with Maximum Sum (hard)
- Solution: Problem Challenge 2: Path with Maximum Sum

14. Pattern: Graphs
11 chapters

- Introduction to Graph
- Graph Representations
- Graph as an Abstract Data Type (ADT)
- Graph Traversal - Depth First Search(DFS)
- Graph Traversal - Breadth First Search (BFS)
- Problem 1: Find if Path Exists in Graph(easy)
- Solution: Find if Path Exists in Graph(easy)
- Problem 2: Number of Provinces (medium)
- Solution: Number of Provinces (medium)
- Problem 3: Minimum Number of Vertices to Reach All Nodes(medium)
- Solution: Minimum Number of Vertices to Reach All Nodes(medium)

15. Pattern: Island (Matrix Traversal)
15 chapters

- Introduction to Island Pattern
- Number of Islands (easy)
- Solution: Number of Islands
- Biggest Island (easy)
- Solution: Biggest Island
- Flood Fill (easy)
- Solution: Flood Fill
- Number of Closed Islands (easy)
- Solution: Number of Closed Islands
- Problem Challenge 1 (easy)
- Solution: Problem Challenge 1
- Problem Challenge 2 (medium)
- Solution: Problem Challenge 2
- Problem Challenge 3 (medium)
- Solution: Problem Challenge 3

16. Pattern: Two Heaps
9 chapters

- Introduction to Two Heaps Pattern
- Find the Median of a Number Stream (medium)
- Solution: Find the Median of a Number Stream
- Sliding Window Median (hard)
- Solution: Sliding Window Median
- Maximize Capital (hard)
- Solution: Maximize Capital
- Problem Challenge 1: Next Interval (hard)
- Solution: Problem Challenge 1: Next Interval

17. Pattern: Subsets
19 chapters

- Introduction to Subsets Pattern
- Subsets (easy)
- Solution: Subsets
- Subsets With Duplicates (easy)
- Solution: Subsets With Duplicates
- Permutations (medium)
- Solution: Permutations
- String Permutations by changing case (medium)
- Solution: String Permutations by changing case
- Balanced Parentheses (hard)
- Solution: Balanced Parentheses
- Unique Generalized Abbreviations (hard)
- Solution: Unique Generalized Abbreviations
- Problem Challenge 1: Evaluate Expression (hard)
- Solution: Problem Challenge 1: Evaluate Expression
- Problem Challenge 2: Structurally Unique Binary Search Trees (hard)
- Solution: Problem Challenge 2: Structurally Unique Binary Search Trees
- Problem Challenge 3: Count of Structurally Unique Binary Search Trees (hard)
- Solution: Problem Challenge 3: Count of Structurally Unique Binary Search Trees

18. Pattern: Modified Binary Search
21 chapters

- Introduction to Modified Binary Search Pattern
- Order-agnostic Binary Search (easy)
- Solution: Order-agnostic Binary Search
- Ceiling of a Number (medium)
- Solution: Ceiling of a Number
- Next Letter (medium)
- Solution: Next Letter
- Number Range (medium)
- Solution: Number Range
- Search in a Sorted Infinite Array (medium)
- Solution: Search in a Sorted Infinite Array
- Minimum Difference Element (medium)
- Solution: Minimum Difference Element
- Bitonic Array Maximum (easy)
- Solution: Bitonic Array Maximum
- Problem Challenge 1: Search Bitonic Array (medium)
- Solution: Problem Challenge 1: Search Bitonic Array
- Problem Challenge 2: Search in Rotated Array (medium)